

ALMA FUNDY NATL PARK, NB, Canada

WMO: 711910

Lat: 45.597N

Lon: 64.953W

Elev: 43

StdP: 100.81

Time Zone: -4.00 (NAA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.0	(c) -16.7	(d) -25.1	(e) 0.4	(f) -18.2	(g) -22.4	(h) 0.5	(i) -14.9	(j) 5.5	(k) -5.2	(l) 5.0	(m) -5.5	(n) 1.8	(o) 350	(p) 0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.0	(c) 24.9	(d) 18.9	(e) 23.6	(f) 18.3	(g) 22.4	(h) 17.9	(i) 20.7	(j) 23.0	(k) 19.8	(l) 22.0	(m) 18.9	(n) 21.1	(o) 1.9	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.7	22.2	18.9	13.8	21.2	18.0	13.0	20.2	60.0	22.8	56.9	22.0	53.9	21.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.5	(b) 4.0	(c) 3.4	(e) -22.6	(f) 28.5	(g) 2.4	(h) 1.7	(i) -24.4	(j) 29.8	(k) -25.8	(l) 30.8	(m) -27.1	(n) 31.8	(o) -28.9	(p) 33.0		
(a) 4.5	(b) 4.0	(c) 3.4	(e) -23.1	(f) 23.1	(g) 2.4	(h) 0.7	(i) -24.8	(j) 23.5	(k) -26.2	(l) 23.9	(m) -27.6	(n) 24.3	(o) -29.3	(p) 24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-5.6	-5.3	-1.5	4.7	10.1	14.3	17.9	18.0	14.8	9.2	3.9	-2.2		
	DBStd	9.31	5.96	5.30	4.66	3.30	2.82	2.54	1.93	1.90	3.19	3.78	4.69	5.54		
	HDD10.0	2096	483	429	356	163	34	2	0	0	2	60	188	379		
	HDD18.3	4329	741	662	614	411	257	122	31	29	110	283	432	638		
	CDD10.0	851	0	0	0	3	36	131	243	247	148	36	6	0		
	CDD18.3	42	0	0	0	0	0	2	16	18	6	1	0	0		
	CDH23.3	142	0	0	0	0	3	12	67	48	12	0	0	0		
	CDH26.7	8	0	0	0	0	0	0	4	2	2	0	0	0		
(14) Wind	WSAvg	1.4	1.6	1.6	1.6	1.5	1.3	1.1	1.0	1.0	1.2	1.3	1.5	1.6		
(15) Precipitation	PrecAvg	1435	130	110	126	107	115	112	98	85	117	140	142	154		
	PrecMax	1871	226	215	286	190	245	258	191	175	308	283	271	319		
	PrecMin	1048	56	31	32	37	36	33	31	9	21	31	54	54		
	PrecStd	187	44	44	53	34	49	60	44	43	67	64	51	56		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	9.3	11.6	17.1	22.5	25.0	27.0	26.2	24.9	21.1	16.8	12.2		
		MCWB	11.1	8.3	6.2	11.6	14.9	18.2	20.2	19.6	19.7	16.8	15.5	11.3		
	2%	DB	7.7	5.8	8.5	14.4	19.2	22.6	25.1	24.6	22.6	18.3	14.8	9.7		
		MCWB	7.0	4.3	5.5	10.1	13.3	16.9	19.0	18.8	18.3	15.5	13.2	8.9		
	5%	DB	5.2	3.6	6.6	12.2	17.1	20.9	23.7	23.4	21.1	16.7	13.0	7.6		
		MCWB	4.1	2.2	4.0	8.9	12.1	15.5	18.4	18.4	17.4	14.6	11.7	6.6		
	10%	DB	2.6	2.1	4.7	10.3	15.2	19.4	22.2	22.2	19.8	15.4	11.2	5.7		
		MCWB	1.6	1.0	2.5	6.9	11.2	15.3	18.0	18.1	16.9	13.3	9.8	4.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	8.3	8.3	14.4	17.0	20.2	22.2	21.9	21.3	18.7	15.9	11.8		
		MCDB	10.7	9.2	9.7	15.4	19.5	22.6	25.5	24.2	23.6	19.8	16.5	12.3		
	2%	WB	6.5	4.5	6.4	11.7	14.8	18.6	20.7	20.6	19.7	16.4	13.7	9.0		
		MCDB	7.0	5.4	7.9	12.9	17.4	20.9	23.0	23.0	21.4	17.6	14.3	9.5		
	5%	WB	3.8	2.4	4.4	9.4	13.3	17.3	19.8	19.7	18.6	14.9	11.8	6.8		
		MCDB	4.6	3.3	5.9	11.3	15.9	19.5	22.2	21.9	20.2	16.4	12.6	7.4		
	10%	WB	1.5	1.0	2.7	7.5	12.0	16.0	18.8	18.7	17.5	13.6	9.8	4.7		
		MCDB	2.2	1.7	4.4	9.7	14.4	18.3	21.2	21.1	19.3	14.9	11.0	5.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.0	8.7	8.7	9.6	9.1	9.4	9.0	9.2	8.4	7.7	7.7		
		MCDBR	11.4	9.2	10.0	12.1	13.2	12.3	11.7	10.7	9.9	9.3	9.3	9.3		
		MCWBR	10.9	8.0	7.6	8.4	7.9	7.3	6.8	6.0	6.2	7.0	8.4	9.0		
		MCWBR	10.6	9.1	9.3	10.4	10.9	9.9	9.8	8.5	8.4	8.1	8.4	9.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.290	0.289	0.323	0.350	0.373	0.399	0.428	0.397	0.363	0.331	0.310	0.290		
		taud	2.418	2.429	2.406	2.382	2.361	2.335	2.290	2.372	2.448	2.534	2.504	2.470		
		Ebn at Noon	821	900	909	910	894	869	837	852	855	841	795	785		
		Edh at Noon	71	85	101	113	119	123	127	113	96	77	66	61		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.45	2.29	3.45	4.30	4.99	5.40	5.54	4.95	3.94	2.51	1.41	1.09		
		RadStd	0.13	0.17	0.22	0.40	0.52	0.45	0.37	0.34	0.30	0.26	0.15	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	+48	N/A

Nomenclature: See separate page